ACC3018 Applied Analytics Capstone

Seminar 3

Agenda

- Endogeneity
- Sources of endogeneity
    - Omitted variable
    - Simultaneous equation
    - Selection bias
    - Measurement error
- Instrumental Variables
- 2SOLS (2SLS)

Motivation

- We consider linear models of the form:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- We compute estimates $\hat{\beta}_1$ to guide choices based on:

$$\Delta y_i = \hat{\beta}_1 \Delta x_i$$

- For this information to be useful it is important that $\hat{\beta}_1 \approx \beta_1$
- Under the classical assumptions we can guarantee:
  - Unbiasedness $\mathbf{E}\ \hat{\beta}_1 = \beta_1$
  - Consistency $\hat{\beta}_1 \xrightarrow[p]{} \beta_1$

Endogeneity

- One of the crucial assumptions is orthogonality
- x_i is exogenous when it is uncorrelated with the error term:

$$\mathbf{E}[\ x_i \varepsilon_i\ ] = 0$$

  Otherwise it is endogenous

- Stronger notions of exogeneity require:
  - ε to be independent from x
  - or conditional mean independence $\mathbf{E}[\ \varepsilon|x\ ] = 0$
- Exogeneity guarantees that the things that are not accounted for (ε_i), do not interfere with the estimation of β_1

Endogeneity bias

- Recall (see slides 3) that we can write:

$$\hat{\beta}_1 = \frac{(x_i - \bar{x})(y_i - \bar{y})}{(x_i - \bar{x})^2} = \beta_1 + \frac{(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{(x_i - \bar{x})^2}$$

- Which implies that:

$$\mathbf{E}\left[\hat{\beta}_1\right] = \beta_1 + \mathbf{E}\left[\frac{(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{(x_i - \bar{x})^2}\right] = 0$$

- And $\hat{\beta}_1$ is an unbiased estimator of β_1 only if:

$$\mathbf{E}\left[\frac{(x_i - \bar{x})(\varepsilon_i - \bar{\varepsilon})}{(x_i - \bar{x})^2}\right] = 0$$

Why doe we care?

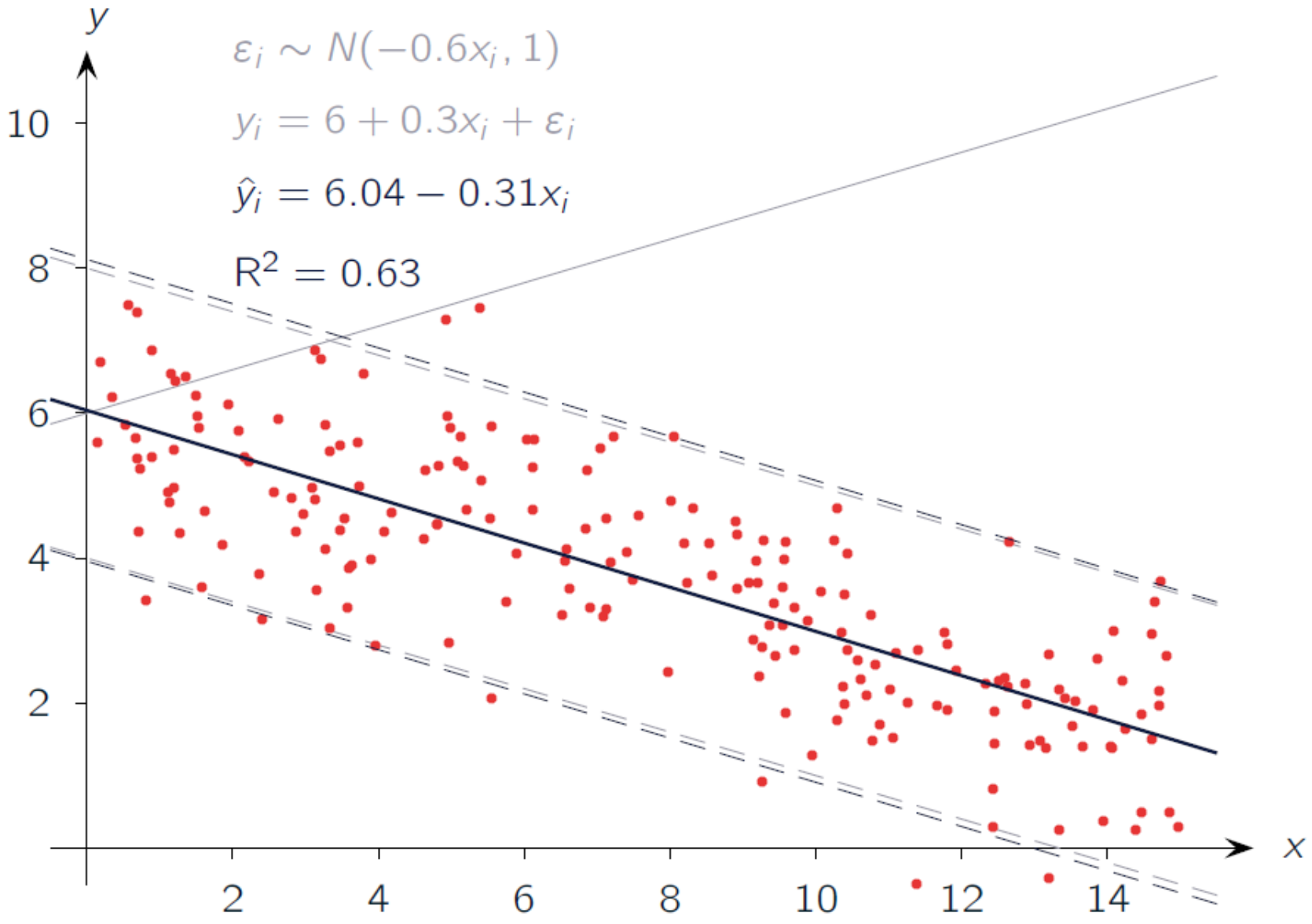
- Orthogonality is the most important and delicate assumption

- Failures of other assumptions can be tested (to some degree)
- Data from a model with a high degree of endogeneity can look completely normal
- Endogeneity can only be established/assumed through common sense/theory

- If other assumptions fail, we can still estimate β_1 consistently, and we can make inference with minor adjustments
- **Endogeneity does not allow to estimate β_1 consistently**
- **If there is strong endogeneity bias, our estimated models can be poor descriptions of reality**

Example: correlation between x and ε

Estimated model



Example: correlation between x and ε

The real problem

- We could define $\eta_i = \varepsilon_i + 0.6x_i$ and rewrite the model as:

$$
\begin{aligned}
y_i &= 6 + 0.3x_i + \varepsilon_i \\
&= 6 + 0.3x_i + (\eta_i - 0.6x_i) = 6 - 0.3x_i + \eta_i
\end{aligned}
$$

- Notice that $\eta_i \sim N(0, 1)$ and $\mathbf{E}[\, x_i \eta_i \,] = 0$
- All classical assumptions are satisfied!

- Why is endogeneity so important then?
  - x_i could be police officers, y_i could be crime rate and ε_i could be demographics
  - A policy that changes x_i may have no effect on ε_i

- Endogeneity is important when we can influence x_i but not ε_i

[0]

1 Endogeneity

2 Sources of endogeneity
Omitted variables
Simultaneous equations
Selection bias
Measurement error

3 Instrumental variables

4 Properties of 2SOLS

Omitted variables

- The most common source of endogeneity is omitted variables
- One of the reason for having an error term ε_i , is because we cannot account for all determinants of y_i
- For our estimates to be consistent, the effect of x_i must not depend on those things that we omit

- Suppose data comes from (with all classical assumptions satisfied):

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \eta_i$$

- And instead we estimate:

$$y_i = \beta_0 + \beta_1 x_{1i} + \varepsilon_i$$

- We omit x_2 (often because we cannot observe it)

Omitted variable bias

- After some algebra:

$$\mathbb{E}\left[\hat{\beta}_1\right] = \beta_1 + \beta_2 \mathbb{E}\left[\frac{\sum(x_{1i}-\bar{x}_1)(x_{2i}-\bar{x}_2)}{\sum(x_{1i}-\bar{x}_1)^2}\right] + \cancelto{0}{\mathbb{E}\left[\frac{\sum(x_{1i}-\bar{x}_1)(\eta_i-\bar{\eta})}{\sum(x_{1i}-\bar{x}_1)^2}\right]}$$

$$= \beta_1 + \beta_2 \mathbb{E}\left[\frac{\sum(x_{1i}-\bar{x}_1)(x_{2i}-\bar{x}_2)}{\sum(x_{1i}-\bar{x}_1)^2}\right]$$

- The term in the expectation is the OLS estimator of α_1 in the model:

$$x_{2i} = \alpha_0 + \alpha_1 x_{1i} + \varphi_i$$

- Therefore we have that:

$$\mathbb{E}\left[\hat{\beta}_1\right] = \beta_1 + \beta_2 \mathbb{E}\left[\hat{\alpha}_1\right] = \beta_1 + \underbrace{\beta_2 \alpha_1}_{\text{bias}}$$

- The OLS estimator of β_1 will be consistent only if there is no bias, i.e.
  - If the omitted variable has no linear relation with y_i ($\beta_2 = 0$)
  - Or if the omitted variable has no linear relation with x_i ($\alpha_1 = 0$)

Example: the returns of schooling

Omitted variable

- Many people are interested in estimating the effect of additional years of schooling (EDU) on earnings (WAGE)
- Typically, people use models of the form

$$\text{WAGE}_i = \beta_0 + \beta_1 \text{ EDU}_i + \gamma \mathbf{x}_i + \varepsilon_i$$

  where $\mathbf{x}_i$ represents a vector of control variables (age, gender, parent's wealth, parent's education, . . .)
- This usually involve some endogeneity because of unobserved characteristics such as innate talents or innate productivity (SKILL)

Example: the returns of schooling

Skill and education

- One would expect that people with stronger ability are prone to stay in school longer
- We would expect a positive relationship between EDU and SKILL, i.e. $\alpha_1 > 0$ in the model

$$\mathrm{SKILL}_i = \alpha_0 + \alpha_1 \mathrm{EDU}_i + \gamma \mathbf{x}_i + \varepsilon_i$$

Example: the returns of schooling

Skill and earnings

- One would expect that more productive people have higher earnings controlling for other factors
- We would expect a positive relationship between WAGE and SKILL, i.e. $\beta_2 > 0$ in the model

$$\text{WAGE}_i = \beta_0 + \beta_1 \text{ EDU}_i + \beta_2 \text{ SKILL}_i + \gamma \mathbf{x}_i + \varepsilon_i$$

Example: the returns of schooling

Omitted variable bias

- From the previous analysis, if we estimate the incomplete model

$$\text{WAGE}_i = \beta_0 + \beta_1 \text{ EDU}_i + \gamma \mathbf{x}_i + \varepsilon_i$$

- We should expect a positive bias:

$$\mathbf{E}\left[\hat{\beta}_1\right] = \beta_1 + \alpha_1 \beta_2 > \beta_1$$

- That is, we would **overestimate** the effect of education
- Since innate skills are not affected by schooling, the direct of β_1 can result in poor policy recommendations

Example: GDP and LifeExp

Omitted variable

```
> m1<-lm(lifeExp ~ gdpPercap, data = gapminder)
> summary(m1)

Call:
lm(formula = lifeExp ~ gdpPercap, data = gapminder)

Residuals:
    Min      1Q  Median      3Q     Max 
-82.754  -7.758   2.176   8.225  18.426 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 5.396e+01  3.150e-01  171.29   <2e-16 ***
gdpPercap   7.649e-04  2.579e-05   29.66   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 10.49 on 1702 degrees of freedom
Multiple R-squared:  0.3407,	Adjusted R-squared:  0.3403 
F-statistic: 879.6 on 1 and 1702 DF,  p-value: < 2.2e-16
```

gdpPercap is significant

Example: GDP and LifeExp

Omitted variable

```
> m2<-lm(lifeExp ~ gdpPercap + continent + year, data = gapminder)
> summary(m2)

Call:
lm(formula = lifeExp ~ gdpPercap + continent + year, data = gapminder)

Residuals:
     Min       1Q   Median       3Q      Max 
-28.4264  -4.0725   0.2154   4.4853  19.9977 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)      -5.207e+02  1.979e+01  -26.31   <2e-16 ***
gdpPercap         2.967e-04  1.996e-05   14.87   <2e-16 ***
continentAmericas 1.433e+01  4.936e-01   29.03   <2e-16 ***
continentAsia     9.506e+00  4.567e-01   20.81   <2e-16 ***
continentEurope   1.940e+01  5.173e-01   37.49   <2e-16 ***
continentOceania  2.059e+01  1.469e+00   14.01   <2e-16 ***
year              2.874e-01  1.000e-02   28.73   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.884 on 1697 degrees of freedom
Multiple R-squared:  0.717,	Adjusted R-squared:  0.716 
F-statistic: 716.6 on 6 and 1697 DF,  p-value: < 2.2e-16
```

gdpPercap is significant after controlling for continent fixed effects and year

Example: GDP and LifeExp

Omitted variable

```
> m3<-lm(formula = lifeExp ~ gdpPercap + year + continent + gdpPercap:year +
+     gdpPercap:continent + year:continent + gdpPercap:year:continent,
+     data = gapminder)
> summary(m3)

Call:
lm(formula = lifeExp ~ gdpPercap + year + continent + gdpPercap:year +
    gdpPercap:continent + year:continent + gdpPercap:year:continent,
    data = gapminder)

Residuals:
     Min       1Q   Median       3Q      Max
-26.7123  -3.2453   0.1535   3.4204  17.7379

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                       -4.241e+02  3.777e+01 -11.229  < 2e-16 ***
gdpPercap                         -2.168e-02  1.226e-02  -1.768  0.07721 .
year                               2.377e-01  1.909e-02  12.457  < 2e-16 ***
continentAmericas                 -4.065e+02  7.317e+01  -5.556 3.20e-08 ***
continentAsia                     -2.258e+02  5.723e+01  -3.946 8.27e-05 ***
continentEurope                    7.212e+01  7.849e+01   0.919  0.35830
continentOceania                   3.047e+02  4.846e+02   0.629  0.52956
gdpPercap:year                     1.149e-05  6.176e-06   1.860  0.06308 .
gdpPercap:continentAmericas        7.174e-02  1.445e-02   4.963 7.63e-07 ***
gdpPercap:continentAsia            7.051e-03  1.251e-02   0.563  0.57318
gdpPercap:continentEurope          4.049e-02  1.317e-02   3.075  0.00214 **
gdpPercap:continentOceania         1.275e-02  3.527e-02   0.362  0.71768
year:continentAmericas             2.119e-01  3.694e-02   5.736 1.15e-08 ***
year:continentAsia                 1.196e-01  2.894e-02   4.133 3.75e-05 ***
year:continentEurope              -2.636e-02  3.970e-02  -0.664  0.50684
year:continentOceania             -1.408e-01  2.461e-01  -0.572  0.56739
gdpPercap:year:continentAmericas -3.634e-05  7.275e-06  -4.996 6.46e-07 ***
gdpPercap:year:continentAsia     -3.945e-06  6.305e-06  -0.626  0.53163
gdpPercap:year:continentEurope   -2.076e-05  6.629e-06  -3.133  0.00176 **
gdpPercap:year:continentOceania  -6.939e-06  1.757e-05  -0.395  0.69289
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.307 on 1684 degrees of freedom
Multiple R-squared:  0.7643,    Adjusted R-squared:  0.7616
```

- gdpPercap is less significant after controlling for continent fixed effects and year and their interactions
- gdpPercap has different effect on life expectancy when we study different continents
- gdpPercap is the only variable in m1, which suffers from endogeneity issues

Simultaneous equations

- This occurs when the independent and dependent variables influence each other in a circular or bidirectional manner.
- Suppose that data is generated according to:

$$y_{1i} = \beta_0 + \beta_1 y_{2i} + \gamma x_i + \varepsilon_i$$
$$y_{2i} = \alpha_0 + \alpha_1 y_{1i} + \theta z_i + \eta_i$$

- Notice that a change in ε_i leads to a change in y_{1i}, which in turns leads to a change in y_{2i}
- Hence ε_i and y_{2i} are correlated!
- For example, education level and income may have a circular relationship, where education influences income, but income can also influence the ability to get more education.

Example: welfare and political stability

- Civil wars are still common in present days.
- Understainding which factors affect political stability, may help to prevent them.

- One may think that an important factor in predicting conflicts (WAR), is the state of the economy (GDP).
- In prosperous times, people are less likely to be discontent or antagonize each other.

- Estimating the effect of GDP on WAR may be hard, because there may be backward causality
  - Civil wars have devastating effects for the economy
  - Instability makes a country less attractive for investors
- Hence we can expect that WAR also affects GDP

Selection bias

- If the sample is not randomly selected, the relationship between the predictor variable and the dependent variable may be distorted.
- For example, if a study only includes people who have a specific characteristic, the results may not be representative of the broader population.
- Suppose we want to estimate the effect of a treatment denoted by a dummy variable d:

$$y_i = \beta_0 + \beta_1 d_i + \varepsilon_i$$

- where

$$\mathbf{E}[\, y_i | d_i = 1 \,] = \beta_0 + \beta_1 + \mathbf{E}[\, \varepsilon_i | d_i = 1 \,]; \text{ and}$$
$$\mathbf{E}[\, y_i | d_i = 0 \,] = \beta_0 + \mathbf{E}[\, \varepsilon_i | d_i = 0 \,]$$

- Therefore

$$\mathbf{E}\ \hat{\beta}_1 = \mathbf{E}[\, y_i | d_i = 1 \,] - \mathbf{E}[\, y_i | d_i = 0 \,]$$
$$= \beta_1 + \mathbf{E}[\, \varepsilon_i | d_i = 1 \,] - \mathbf{E}[\, \varepsilon_i | d_i = 0 \,]$$

- We obtain consistent estimates only if $\mathbf{E}[\, \varepsilon_i | d_i = 1 \,] = \mathbf{E}[\, \varepsilon_i | d_i = 0 \,]$
- That is, if the selection criteria for the treatment is orthogonal to ε

Example: the returns of military service

- It is important to understand what are the long-term consequences of military service (MS) for people who will not pursue military careers
- In particular one could be concerned about the effect on income (WAGE)

$$\text{WAGE}_i = \beta_0 + \beta_1 \text{MS}_i + \beta_2 \text{MS}_i \cdot \text{AGE}_i + \beta_3 \text{MS}_i \cdot \text{AGE}^2_i + \gamma \mathbf{x}_i + \varepsilon_i$$

- A potential problem is that the choice to participate in voluntary service is endogenous
- Candidates with less attractive outside options have lower opportunity cost and are more likely to participate
- This same candidates are likely to have higher wages *caeteris paribus*
- Hence it is likely that:

$$\mathbb{E}[\varepsilon_i | \text{MS}_i = 1] \neq \mathbb{E}[\varepsilon_i | \text{MS}_i = 0]$$

Measurement error

- If the values of a predictor variable are not accurately measured, the observed variable will be correlated with the error term.
- This can lead to biased estimates, as the model is effectively estimating the relationship between the measured variable and the dependent variable, not the true relationship between the underlying construct and the dependent variable
- The measurement of economic activity may be susceptible of measurement error
- Consider the model: $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

- Suppose that instead of observing x, we observe $x^* = x + \mu$
- $x = x^* - \mu$
- Hence the model that we actually estimate is: $y_i = \beta_0 + \beta_2 x^* + \eta$
- In many cases, it is sensible to assume that the measurement error is correlated with the measured variable

Example: health and wealth

- Wealthier people tend to be healthier
- Better nutrition, better medial services . . .

- Wealth data is often self reported
- Self-reports of wealth usually carry some measurement error
  - Most people don't know the exact value of their assets and income
  - Some people may want to distort information for privacy concerns

- Both effects tend to be higher for richer people
- Hence we may have $\mathbf{E}[\, x_i \mu_i \,] < 0$

[0]

1 Endogeneity

2 Sources of endogeneity
Omitted variables
Simultaneous equations
Selection bias
Measurement error

3 Instrumental variables

4 Properties of 2SOLS

The problem

- We wish to estimate the effect of x on y, namely β_1
- The problem with endogeneity is that:

  We cannot disentangle the variation of x from variation of other factors (ε) which may also affect y

- For example:
  - If we only observe prices and quantities, we cannot tell whether increases in quantities are due to increases in prices or to increases in demand
  - If we only observe earnings and years of schooling, we do not know if high earnings are due to education or skill

Natural experiments

- If we could observe variations of x which are independent from ε, we could then estimate β_1

- This is exactly what happens in controlled experiments in which the researcher varies x keeping everything else constant
- Controlled experiments are rarely possible in realistic economic scenarios
- Laboratory experiments in Economics are often questioned for lack of external validity

- An alternative is to look for natural experiments which affect x but nothing else
- The idea of natural experiments is materialized in instrumental variables

Instrumental variables

- An instrumental variable (IV) is a variable that is related to x but is otherwise independent from y
  - It affects y but only through x
  - In particular it is independent (orthogonal) to ε
- An example of an intrumental variable are the manipulations of a researcher in an experimental setting
- Instead of considering all the variation of x, focus on the variation explained by z
- This variation is exogenous!
- If this variation is large enough, we can use it to identify β_1
- In that sense IVs can serve as instruments to identify the effect of x on y

Two stage ordinary least squares

Setup

- To study the effect of education on wages, there's a potential endogeneity problem years of schooling captures only what is taught in school or college. And it also leaves out aspects such as how well the person has grasped the material, their knowledge of topics outside of the curriculum and so on, all of which are left unobserved
- To address this, you might use parental education levels as an **instrument** (they influence the child's education but are not directly correlated with the child's wages).
- Frist stage:

$$\begin{aligned}\boldsymbol{education} &= \gamma_1 + \gamma_2 * \boldsymbol{age} + \gamma_3 * \boldsymbol{experience} + \gamma_4 * \boldsymbol{college} \\ &+ \gamma_5 * \boldsymbol{unemp} + \gamma_6 * \boldsymbol{city} \\ &+ \gamma_7 * \boldsymbol{meducation} + \gamma_8 * \boldsymbol{feducation} + \boldsymbol{v}\end{aligned}$$

- Estimate the education:

$$\begin{aligned}\widehat{\boldsymbol{education}} &= \hat{\gamma}_1 + \hat{\gamma}_2 * \boldsymbol{age} + \hat{\gamma}_3 * \boldsymbol{experience} + \hat{\gamma}_4 * \boldsymbol{college} \\ &+ \hat{\gamma}_5 * \boldsymbol{unemp} + \hat{\gamma}_6 * \boldsymbol{city} \\ &+ \hat{\gamma}_7 * \boldsymbol{meducation} + \hat{\gamma}_8 * \boldsymbol{feducation}\end{aligned}$$

- Second stage:

$$\begin{aligned}ln(\boldsymbol{wage}) &= \beta_1 + \beta_2 * \boldsymbol{age} + \beta_3 * \boldsymbol{experience} + \beta_4 * \boldsymbol{college} \\ &+ \beta_5 * \boldsymbol{unemp} + \beta_6 * \boldsymbol{city} + \beta_7 * \widehat{\boldsymbol{education}} + \boldsymbol{\epsilon}\end{aligned}$$

Two stage ordinary least squares

Setup

- Suppose that you want to estimate:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

- However you are afraid that x may be endogenous

- To fix this problem, you use z as an instrument
- z should be related to x and independent from ε

- Instead of considering all the variation in x we can focus by the variation of x which can be explained by z
- We do this using two stage ordinary least squares (2SOLS)

Example: market equilibrium

demand and supply shifters

Demand	Supply
Population	Number of competitors
Income	Technological shocks
Preferences (ads,fads)	Prices of inputs
Expectations	Expectations
Prices of substitutes/complements	Prices of alternative goods

Two stage ordinary least squares

First stage

- To capture the variation of x which can be explained by z, consider the model

$$x_i = \alpha_0 + \alpha_1 z_i + \eta_i$$

- The first step of 2SOLS is to run regular OLS on this model to obtain estimates $\hat{\alpha}_0$ and $\hat{\alpha}_1$

- We then use these estimates to compute predicted values:

$$\hat{x}_i = \hat{\alpha}_0 + \hat{\alpha}_1 z_i$$

which are independent from ε_i!!

Two stage ordinary least squares

Second stage

- The predictions $\hat{x}_i$ contain all the variation of x which can be explained by z
- Assuming that η and z are exogenous, then so is $\hat{x}$

- We can use $\hat{x}$ to estimate the effect of x over y by running OLS on

$$y_i = \beta_0 + \beta_1 \hat{x}_i + \varphi_i$$

- The estimators obtained in the second stage are the IV estimators of β_0 and β_1
- They are often denoted by $\hat{\beta}_0^{\text{IV}}$ and $\hat{\beta}_1^{\text{IV}}$

Valid instruments

- Which variables can serve as instruments?

- A first obvious requirement is that it must be something which can be measured (data should be available)

- A good instrument should explain some of the variation of x
- The slope coefficient in the first stage model should be significant
- The R^2 coefficient should be high enough!

- A good instrument should be exogenous (orthogonal to ε)
- Unfortunately, endogeneity cannot be tested in general
- The endogeneity of the instruments has to be established via common sense/theory

Examples of instrumental variables

Response	Regressor	Eondogeneity	Instruments
earnings	schooling	omitted	quarter of birth school construction proximity to college
newborn weight	smoking	omitted	random assignement state taxes
demand supply	price price	simultaneity simultaneity	supply shifters demand shifters
crime	police surveillance	simultaneity	electoral cycles Papal trajectory
conflict	GDP	simultaneity	rainfall
labor supply	fertility	simultaneity	gender composition of children
earnings	military service	selection	draft lottery

Seminar Activity

- Find a paper (Accounting and Finance) using google scholar where two-stage regression is performed. Cite the paper in APA format.
- Identify the IV and reason for 2-Stage Least Squares (2SLS) Estimation.
- Identify the model, regression results, and discussion of the regression results.
- Is there any limitation and future research? If so, identify them.